

Introduction

Surface-Enhanced Raman Spectroscopy (SERS) is a powerful vibrational spectroscopy technique that enhances weak Raman signals by several orders of magnitude when molecules are adsorbed onto nanostructured metallic surfaces, typically gold or silver. This enables highly sensitive, label-free molecular fingerprinting of biomolecules, making SERS well suited for biomedical diagnostics. However, the high dimensionality and variability of SERS spectra complicate data interpretation and hinder clinical translation. We hypothesize that machine learning (ML) approaches could address these challenges by extracting meaningful spectral features, reducing noise, and improving classification accuracy. In this study, we employed a range of ML algorithms to classify SERS spectra from seven amino acids biomineralized onto AuNPs. Using PCA, LDA, and PCA+LDA for dimensionality reduction with supervised classification models, we aim to develop a robust computational framework for AI-driven, non-invasive, label-free SERS diagnostics in personalized medicine.

Our Approach

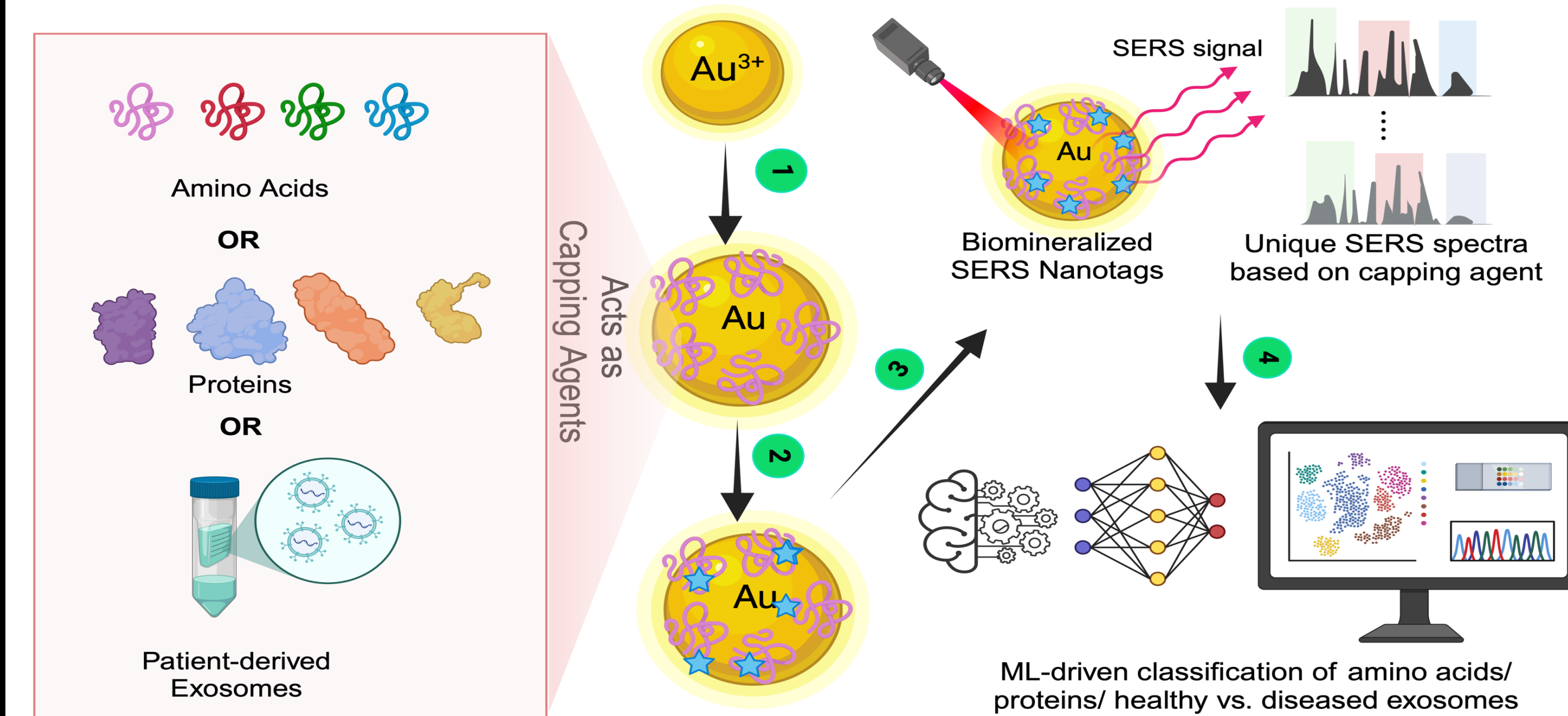


Figure 1: Biomaterialization-driven SERS gold nanotags enable machine learning-based disease diagnostics through unique spectral signatures.

Machine Learning Workflow for SERS Data Analysis

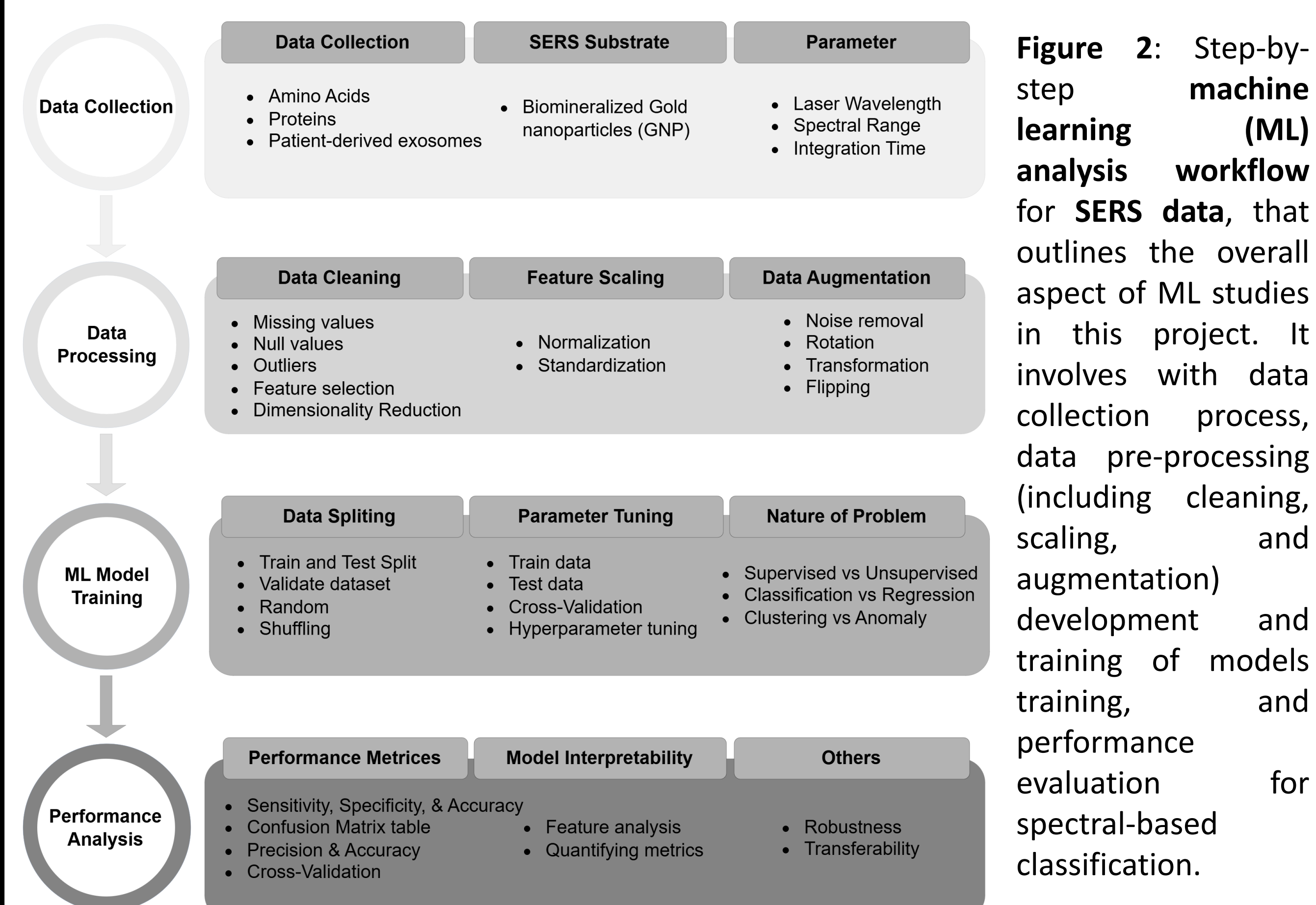


Figure 2: Step-by-step machine learning (ML) analysis workflow for SERS data, that outlines the overall aspect of ML studies in this project. It involves with data collection process, data pre-processing (including cleaning, scaling, and augmentation) development and training of models training, and performance evaluation for spectral-based classification.

Physicochemical Characterization of Biomineralized SERS Nanotags

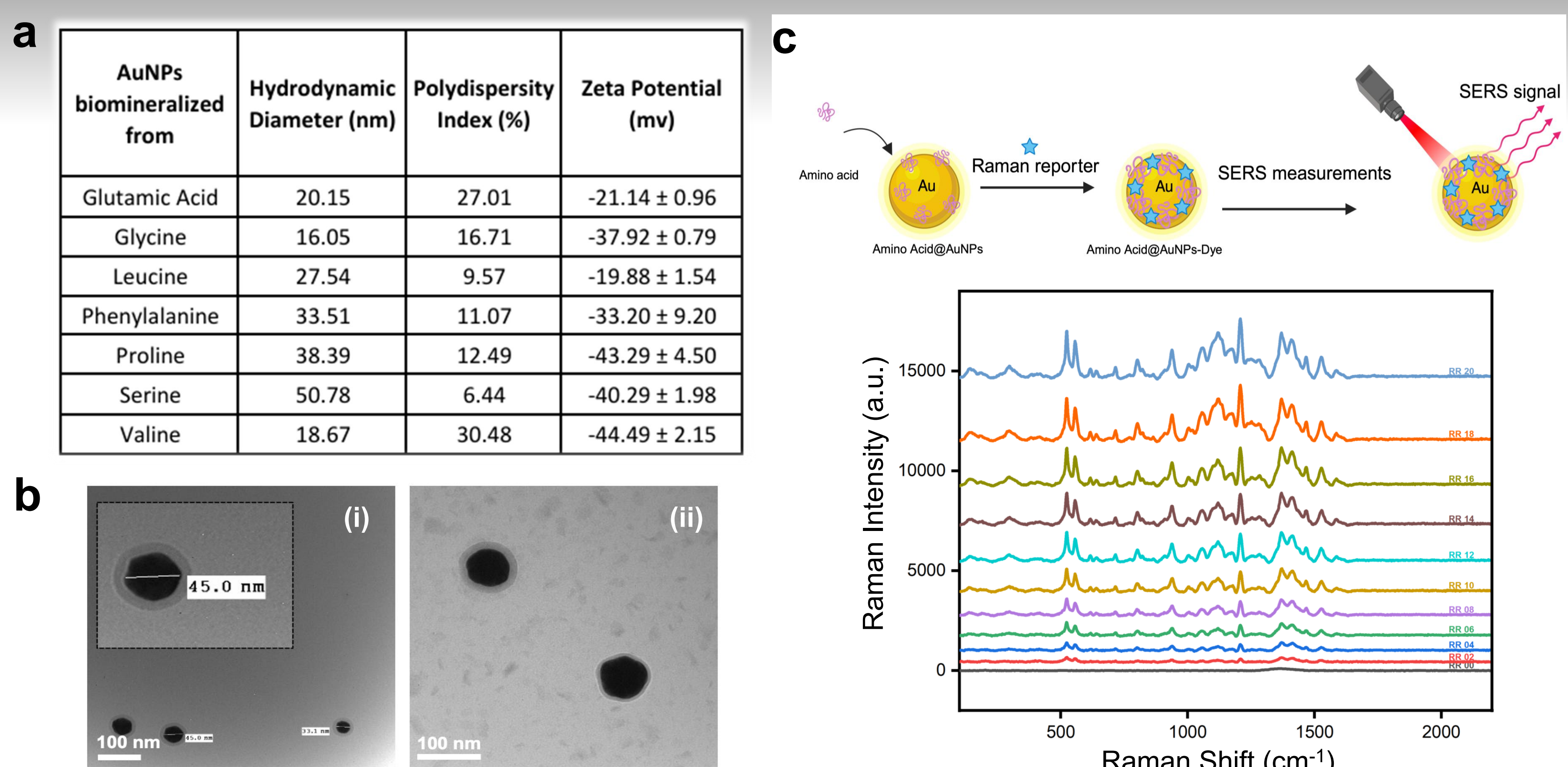


Figure 3: (a) Hydrodynamic diameter, polydispersity index, and zeta potential values of biomineralized AuNPs synthesized using different alpha amino acids. (b) TEM images of Leucine (i) and Serine (ii) – capped AuNPs. (c) SERS spectra showing variations in SERS signal intensity of Valine@SERS nanotags with different concentrations of a Raman reporter.

Machine Learning Based Separation of Distinct Amino Acids

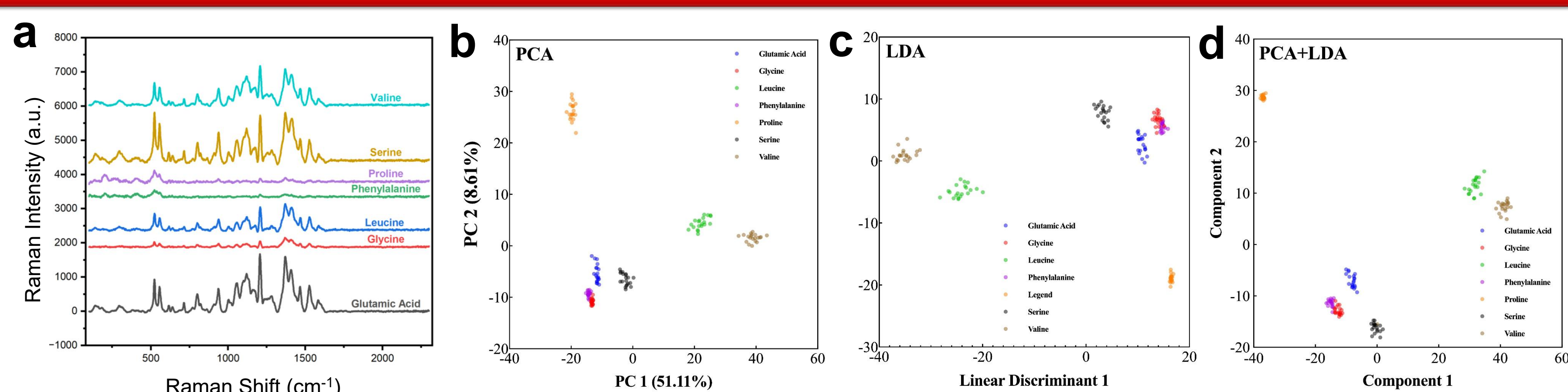


Figure 4: (a) Comparative SERS spectra for different Raman reporters at a fixed concentration. (b) Principal component analysis (PCA) score plot visualizing the distribution of SERS spectra from amino acid-biomineralized nanotags, (c) Linear Discriminant Analysis (LDA) score plot visualizing the distribution of SERS spectra from amino acid-biomineralized nanotags, (d) PCA+LDA score plot showing distinct grouping of the nanotags according to amino acid type.

Ratiometric Separation of Two Amino Acids

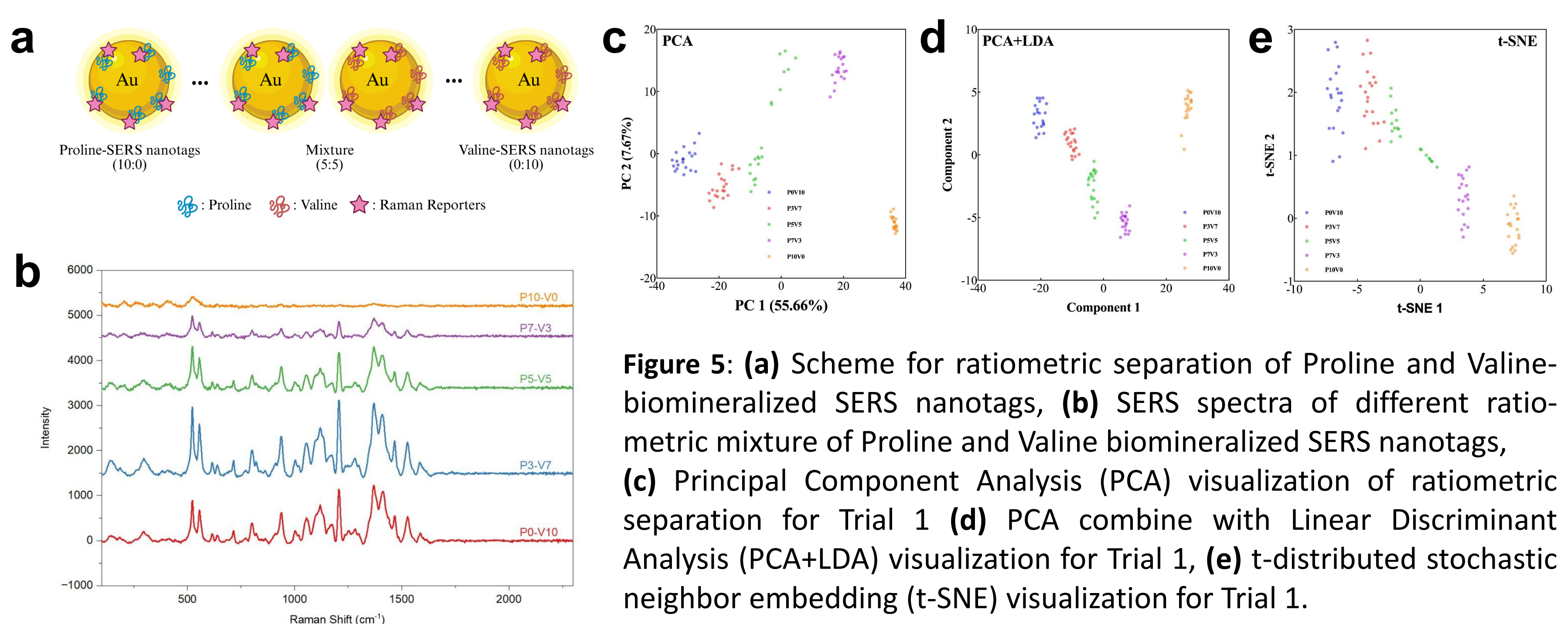


Figure 5: (a) Scheme for ratiometric separation of Proline and Valine-biomineralized SERS nanotags, (b) SERS spectra of different ratiometric mixture of Proline and Valine biomineralized SERS nanotags, (c) Principal Component Analysis (PCA) visualization of ratiometric separation for Trial 1 (d) PCA combine with Linear Discriminant Analysis (PCA+LDA) visualization for Trial 1, (e) t-distributed stochastic neighbor embedding (t-SNE) visualization for Trial 1.

Model Training and Validation Studies

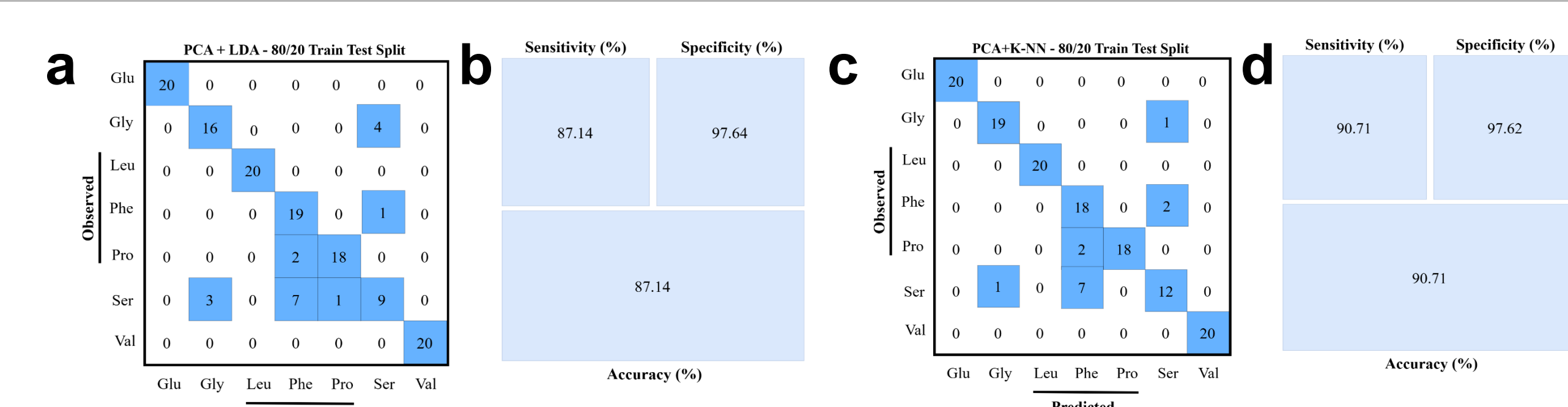


Figure 6: (a, c) Confusion matrices for the PCA-LDA and PCA-KNN model trained with 80/20 train-test splits showing classification accuracy for seven amino acids. (b, d) Corresponding sensitivity (Sen%), specificity (Spe%), and overall accuracy metrics for each model.

Single Blind Studies to Predict Amino Acids

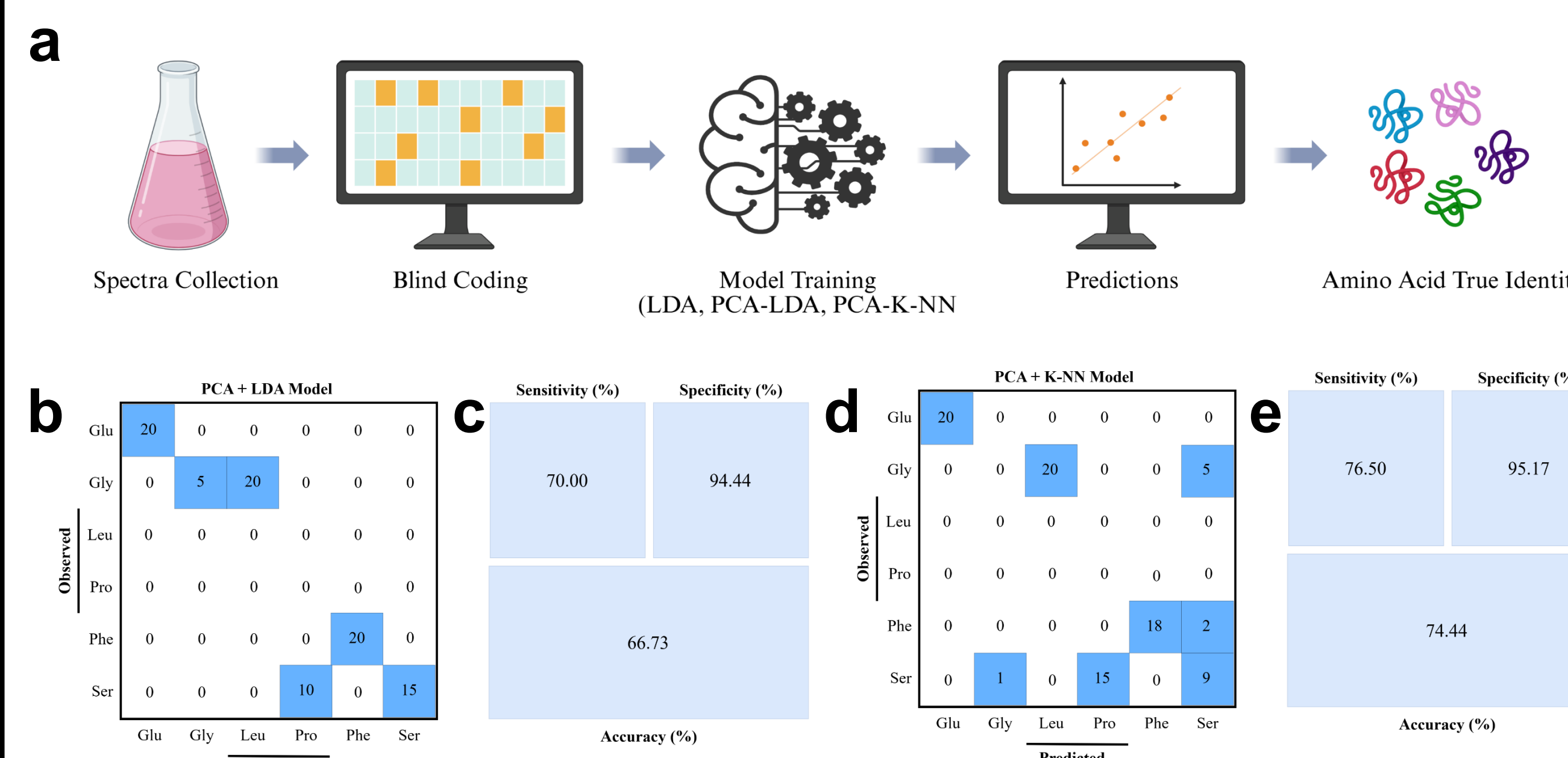


Figure 7: (a) Scheme for single-blind study, (b) Confusion matrix for the PCA+LDA model applied in a single-blind study to classify three unknown amino acids. (c) Sensitivity, specificity, and accuracy metrics for the model's classification performance. (d) Confusion matrix for the PCA+KNN model applied in a single-blind study to classify three unknown amino acids. (e) Sensitivity, specificity, and accuracy metrics for the model's classification performance.

Conclusion and Future Work

- We developed a reliable machine learning pipeline to classify AA-AuNPs SERS spectra with a high degree of accuracy.
- The integration of dimensionality reduction and supervised learning was essential for robust performance on both known and unknown data.
- PCA+K-NN model achieved the highest accuracy (74.44%), offering a scalable and interpretable solution for biomarker identification in unknown samples.
- We intend further explore advanced Machine Learning methods and expand our repertoire to deep learning models like Convolutional Neural Networks (CNNs), to enhance diagnostic robustness and adaptability.
- Future work will extend this approach to more complex biological targets such as proteins, bacterial strains, and patient-derived exosomes which carry disease-related molecular biomarkers.

References

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- [3] I. Srivastava et al., "Biomimetic Surface-Enhanced Raman Scattering Nanoparticles with Improved Dispersibility, Signal Brightness, and Tumor Targeting Functions," *ACS Nano*, vol. 16, no. 5, pp. 8051–8063, Apr. 2022, doi: <https://doi.org/10.1021/acsnano.2c01062>.

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